

Feedforward Filters

General Form $y_t = x_t + ax_{t-1}$ $H(\omega) = 1 + ae^{-i\omega}$ $H(z) = 1 + az^{-1}$ $ H(\omega) = (1 + a^2 + 2a \cos(\omega))^{\frac{1}{2}}$	Inverse Comb $y_t = x_t + ax_{t-L}$ $H(\omega) = 1 + ae^{-i\omega L}$ $H(z) = 1 + az^{-L}$ $ H = 2 \sin(\omega L/2) $...
Zeroes $z^L = -1/a$	Transfer Function $y_t = x_t + ax_{t-1}$ $\mathbf{Y} = \mathbf{X}(1 + az^{-1})$ $\frac{\mathbf{Y}}{\mathbf{X}} = (1 + az^{-1})$ $H(z) = 1 + az^{-1}$...

Phase Response $[R_\omega e^{i\theta\omega}]$ R_ω : MAGNITUDE RESPONSE

$y_t = x_t - 2x_{t-1}$ $H(\omega) = 1 - 2e^{-i\omega} = \frac{e^{i\omega}-2}{e^{i\omega}}$ $ H(\omega) = [5 - 2\cos(\omega)]^{1/2}$ $\theta_\omega = \tan^{-1}(\frac{2\sin(\omega)}{1-2\cos(\omega)})$	$\theta_\omega =$ $\tan^{-1}(\frac{ImaginaryPart}{RealPart})$ $e^{i\theta\omega} = \frac{H(\omega)}{ H(\omega) }$ $\theta_\omega H(\omega) = e^{i\omega}(1 + ae^{-i\omega})$
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Palindromic Phase Response ODD TERMS $H(z) = 2 + z^{-1} - 3z^{-2} + z^{-3} + 2z^{-4}$ $H(\omega) = 2 + e^{-i\omega} - 3e^{-2i\omega} + e^{-3i\omega} + 2e^{-4i\omega}$ $H(\omega) = e^{-i2\omega} [4\cos(2\omega) + 2\cos(\omega) - 3]$ $ H(\omega) = 4\cos(2\omega) + 2\cos(\omega) - 3$ $\theta_\omega = -2\omega$...	Even Terms $y_t = x_t + \frac{1}{3}x_{t-1} + \frac{1}{3}x_{t-2} + x_{t-3}$ $H(\omega) = 1 + \frac{1}{3}e^{-i\omega} + \frac{1}{3}e^{-2i\omega} + e^{-3i\omega}$ $H(\omega) = e^{-3i\frac{\omega}{2}} [\frac{1}{2}e^{3i\frac{\omega}{2}} + \frac{1}{3}e^{i\frac{\omega}{2}} + \frac{1}{3}e^{-i\frac{\omega}{2}} + e^{-3i\frac{\omega}{2}}]$ COLLECT COS WITH $e^{i\frac{\omega}{2}} + e^{-i\frac{\omega}{2}} = 2\cos(\frac{\omega}{2})$
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Phasors $f(t) = e^{i\omega t}$ ω = angular freq (rad/s) Frequency: $f = \frac{\omega}{2\pi}$ Hz Period: $T = \frac{2\pi}{\omega} = \frac{1}{f}$ seconds	Function $f(z) = \alpha z$ rotates by $\arg(\alpha)$, scales by $ \alpha $ Aliasing Sample rate: f_s Hz Nyquist frequency: $f_s/2$ $f_{alias} = f - n f_s $ (choose n to minimize)
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First Order:

$y_t = x_t + R y_{t-1}$ $H(z) = \frac{1}{1-Rz^{-1}} = \frac{z}{z-R}$ • Pole at $z = R$ (stable if $ R < 1$) • Impulse response: $h[n] = R^n u[n]$ • $ H(\omega) = \frac{1}{ 1-Re^{-i\omega} }$	Reson Filter (2-Pole) Poles: $z = Re^{\pm i\theta}$ $H(z) = \frac{1}{1 - 2R \cos \theta \cdot z^{-1} + R^2 z^{-2}}$ Properties: • Peak at $\omega = \theta$ (resonant freq) • Bandwidth: $B \approx 2(1 - R)$ • For narrow band: $R \approx 1 - B/2$ • Peak gain: $\approx \frac{1}{1-R}$ for $R \approx 1$ • Max $ H(\omega) $: $\approx \frac{1}{\sqrt{1-R^2}}$ at resonance • True max: $\cos(\omega_m) = \frac{1+R^2}{2R}$ Implementation: $y_t = x_t + 2R \cos \theta \cdot y_{t-1} - R^2 y_{t-2}$
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Comb Filter

Form: $H(z) = \frac{1}{1-R^L z^{-L}}$ Impulse Response Sum: $\sum_{t=0}^\infty h_t = \frac{1}{1-R^L}$ Properties:
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- Peaks at $\omega = 2\pi k/L$
- Nulls depend on R value
- Creates harmonic series

All-Pass Filter

General Form: $H(z) = \frac{a+z^{-1}}{1+az^{-1}}$ where $ a < 1$ Properties:
• $ H(\omega) = 1$ for all ω • Phase adjustment without magnitude change • Numerator = conjugate reciprocal of denominator • Used for fractional delay

Phase Delay: $\tau(\omega) = \frac{-2 \arctan(\frac{a \sin \omega}{1+a \cos \omega})}{\omega}$

Plucked String (Karplus-Strong)

Components:
• Delay line: L samples • Low-pass: $y = \frac{1}{2}(x_t + x_{t-1})$, adds 0.5 delay • All-pass: fine tuning

Quantization

• Period: $P = f_s/f_0$ samples • Integer delay: $L = \lfloor P \rfloor$ • Fractional: $\delta = P - L$ • All-pass coeff: $a = \frac{1-\delta}{1+\delta}$	Calculations Quantization Uniform Error $[-0.5, 0.5]$: RMS = $\frac{1}{2\sqrt{3}} \approx 0.289$ dB = $20 \log_{10}(A)$
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Unit Circle $e^{i\pi/2} = i, e^{i\pi} = -1,$ $e^{i3\pi/2} = -i, e^{i2\pi} = 1$ $e^{i\pi/3} = \frac{1}{2} + \frac{\sqrt{3}}{2}i,$ $e^{i2\pi/3} = -\frac{1}{2} + \frac{\sqrt{3}}{2}i$ $e^{i\pi/4} = \frac{1+i}{\sqrt{2}},$	$e^{i\pi/6} = \frac{\sqrt{3}}{2} + \frac{1}{2}i$ Complex Arithmetic $z = a + bi = re^{i\theta}$ $r = z = \sqrt{a^2 + b^2}$ $i^2 = -1, i^3 = -i, i^4 = 1,$ $i^5 = i$
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Operations $(a + bi)(c + di) = (ac - bd) + (ad + bc)i$ $\frac{a+bi}{c+di} = \frac{(ac+bd)+(bc-ad)i}{c^2+d^2} (re^{i\theta})^n = r^n e^{in\theta}$ Conjugate: $\bar{z} = a - bi = re^{-i\theta}$ Rotation by θ : multiply by $e^{i\theta}$ Trig Identities

$ 1 + e^{i\omega} = 2 \cos(\omega/2) $ $ 1 - e^{i\omega} = 2 \sin(\omega/2) $ $(a + bi)(a - bi) = a^2 + b^2$	$e^{it} - e^{-it} = 2i \sin(t)$ $e^{it} + e^{-it} = 2 \cos(t)$
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Quadratic from Complex Roots $(x - (a + bi))(x - (a - bi)) = x^2 - 2ax + (a^2 + b^2)$ Cubic Factoring $x^3 + 1 = (x + 1)(x^2 - x + 1)$ $x^3 - 1 = (x - 1)(x^2 + x + 1)$

Taylor Series $e^{ix} = 1 + i\frac{\pi}{2} + \frac{i(\frac{\pi}{2})^3}{3!} + \dots$ $\sum_{k=0}^\infty \frac{x^k}{k!} = e^{ix}$	Product $\prod_{k=1}^n e^{ik\theta} = e^{i\theta n(n+1)/2}$ Geometric Series $\sum_{k=0}^n e^{ik\theta} = \frac{1 - (e^{i\theta})^{n+1}}{1 - e^{i\theta}}$
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+3dB = $\sqrt{2}$ × amplitude
+6dB = 2× amplitude
+20dB = 10× amplitude

DFT & Fourier Basis
Basis Vectors: $u_k[n] = e^{i2\pi kn/N}$
Orthogonality: $u_k \cdot u_j = N\delta_{kj}$

$X[k] = \sum_{n=0}^{N-1} x[n]e^{-i2\pi kn/N}$
$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k]e^{i2\pi kn/N}$
Coefficient: $c_k = X[k]/N$ Inner Product: $\langle u, v \rangle = \sum_n \bar{u}_n v_n$

Square Wave Fourier

Period T , 50% duty cycle:

$$c_k = \begin{cases} 0 & k \text{ even} \\ \frac{4}{k\pi} & k \text{ odd} \end{cases}$$

General Fourier: $c_k = \frac{1}{T} \int_0^T s(t)e^{-i2\pi kt/T} dt$

Filter Combinations

- **Cascade:** $H_{total} = H_1 \cdot H_2$
- **Parallel:** $H_{total} = H_1 + H_2$
- **Feedback:** $H_{total} = \frac{GH}{1-FH}$

Phase Properties

- Linear Phase:**
- Symmetric coefficients
- $\phi(\omega) = -\omega D$ where D = group delay
- No distortion of wave shape

Phase Delay vs Group Delay:

- Phase delay: $\tau_p = -\phi(\omega)/\omega$
- Group delay: $\tau_g = -d\phi/d\omega$

Critical Exam Formulas

Operation	Formula
Complex dot	$(a_1, b_1) \cdot (a_2, b_2) = \bar{a}_1 a_2 + b_1 b_2$
Parseval	$\frac{1}{T} \int s ^2 dt = \sum c_k ^2$
Impulse	$\delta[n] = \{1 \text{ if } n = 0, 0 \text{ else}\}$